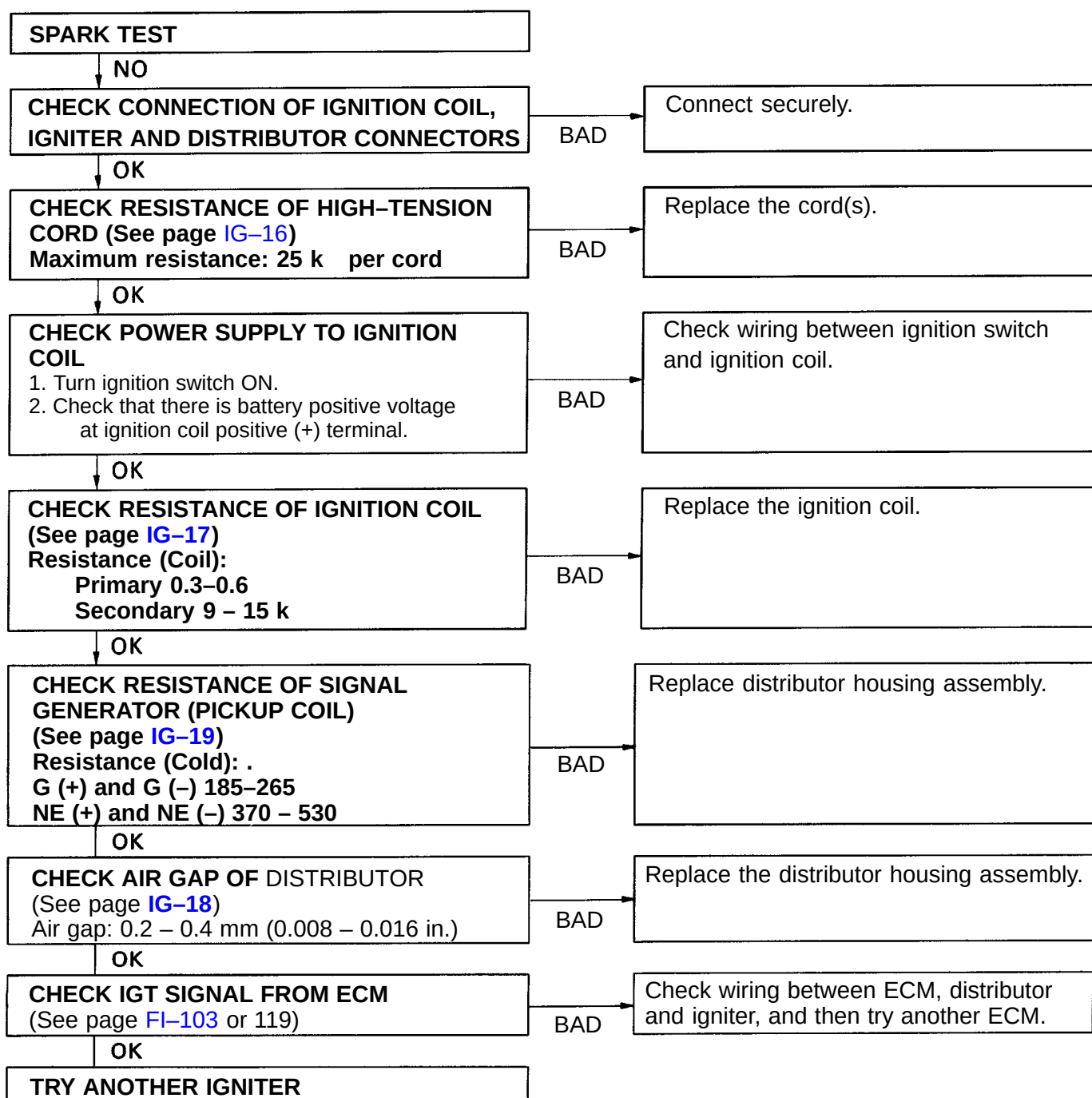


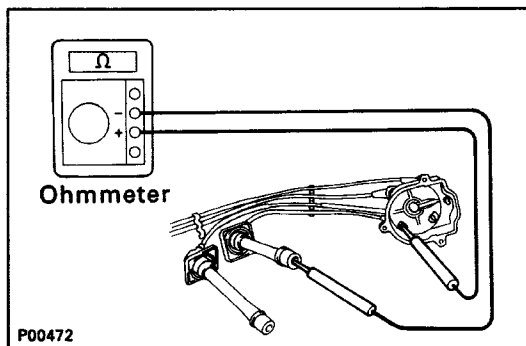
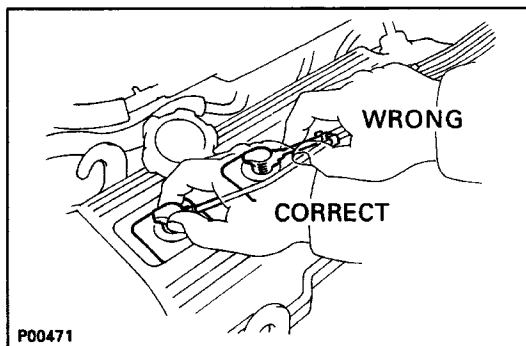
## ON-VEHICLE INSPECTION (5S-FE)

### SPARK TEST

#### CHECK THAT SPARK OCCURS

- (a) Disconnect the high-tension cord from the distributor.  
(See page [IG-16](#))
  - (b) Hold the end about 12.5 mm (0.50 in.) from the body of car.
  - (e) Check if spark occurs while engine is being cranked.  
HINT: To minimize the amount of fuel injected into the cylinders during this test, crank the engine for no more than 1 – 2 seconds at a time.
- If the spark does not occur, perform the test as follows:





## INSPECTION OF HIGH-TENSION CORDS

### 1. DISCONNECT HIGH-TENSION CORDS FROM SPARK PLUGS

Disconnect the high-tension cords at the rubber boot.  
DO NOT pull on the cords.

**NOTICE:** Pulling on or bending the cords may damage the conductor inside.

### 2. DISCONNECT HIGH-TENSION CORD FROM IGNITION COIL (See step 3 on pages IG-30 and 31)

### 3. REMOVE DISTRIBUTOR CAP WITHOUT DISCONNECTING HIGH-TENSION CORDS

### 4. INSPECT HIGH-TENSION CORD RESISTANCE

Using an ohmmeter, measure the resistance without disconnecting the distributor cap.

**Maximum resistance: 25 k  $\Omega$  per cord**

If the resistance is greater than maximum, check the terminals. If necessary, replace the high-tension cord and/or distributor cap.

### 5. REINSTALL DISTRIBUTOR CAP

### 6. RECONNECT HIGH-TENSION CORD TO IGNITION COIL (See step 3 on pages IG-32 and 33)

### 7. RECONNECT HIGH-TENSION CORDS TO SPARK PLUGS

## INSPECTION OF SPARK PLUGS

### NOTICE:

- Never use a wire brush for cleaning.
- Never attempt to adjust the electrode gap on used spark plug.
- Spark plug should be replaced every 100,000 km (60,000 miles).

### 1. DISCONNECT HIGH-TENSION CORDS FROM SPARK PLUGS

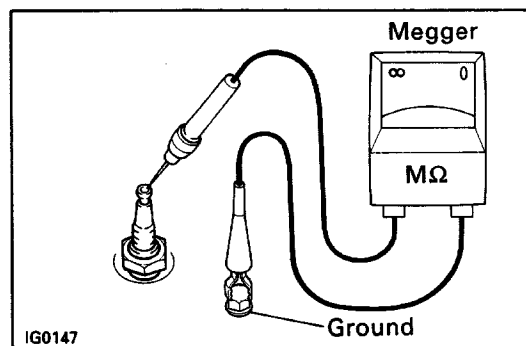
### 2. INSPECT ELECTRODE

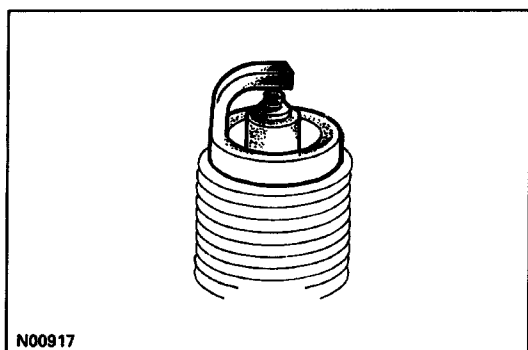
Using a megger (insulation resistance meter), measure the insulation resistance.

**Standard correct insulation resistance:**  
**10 M  $\Omega$  or more**

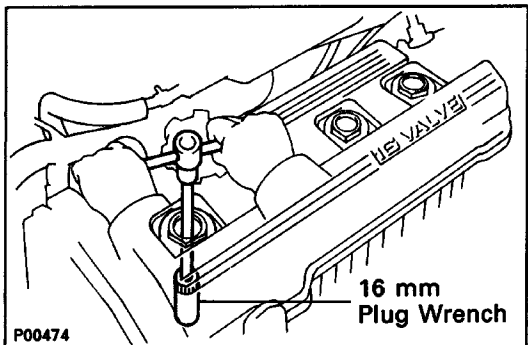
If the resistance is less than specified, proceed to step 4.

**HINT:** If a megger is not available, the following simple method of inspection provides fairly accurate results.

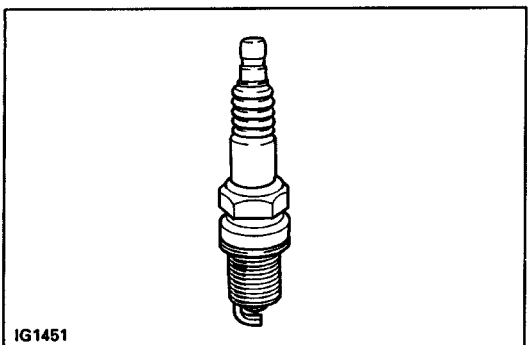


**(Simple Method)**

- (a) Quickly race the engine to 4,000 rpm five times.
- (b) Remove the spark plug. (See step 3)
- (c) Visually check the spark plug.  
If the electrode is dry ... Okay  
If the electrode is wet ... Proceed to step 4
- (d) Reinstall the spark plug.  
(See step 7 on page IG-18)

**3. REMOVE SPARK PLUGS**

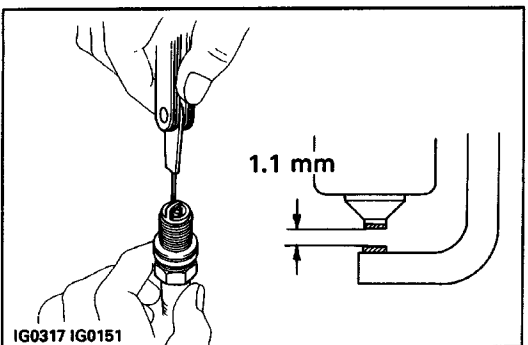
Using a 16 mm plug wrench, remove the spark plug.

**4. VISUALLY INSPECT SPARK PLUGS**

Check the spark plug for thread damage and insulator damage.

If abnormal, replace the spark plug.

**Recommended spark plug:** ND PK20R11  
NGK BKR6EP-11

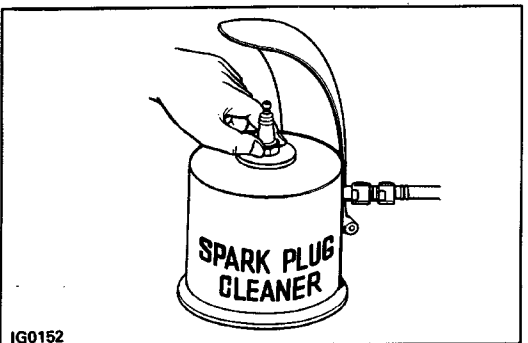
**5. INSPECT ELECTRODE GAP**

**Maximum electrode gap: 1.3 mm (0.051 in.)**

If the gap is greater than maximum, replace the spark plug.

**Correct electrode gap of new spark plug:**  
**1.1 mm (0.043 in.)**

**NOTICE:** If adjusting the gap of a new spark plug, bend only the base of the ground electrode. Do not touch the tip. Never attempt to adjust the gap on the used plug.

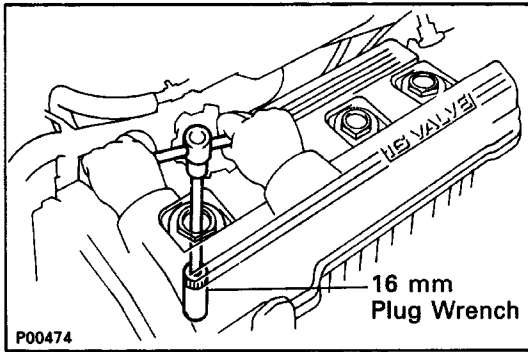
**6. CLEAN SPARK PLUGS**

If the electrode has traces of wet carbon, allow it to dry and then clean with a spark plug cleaner.

**Air pressure:** Below 588 kPa (6 kgf/cm<sup>2</sup>, 85 psi)

**Duration:** 20 seconds or less

**HINT:** If there are traces of oil, remove it with gaso-line before using the spark plug cleaner.



## 7. INSTALL SPARK PLUGS

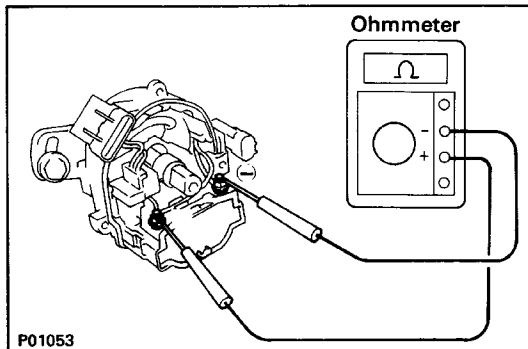
Using a 16 mm plug wrench, install the spark plug.

**Torque: 18 N·m (180 kgf·cm, 13 ft·lbf)**

## 8. RECONNECT HIGH-TENSION CORDS TO SPARK PLUGS

### INSPECTION OF DISTRIBUTOR

1. DISCONNECT DISTRIBUTOR CONNECTORS
2. DISCONNECT DISTRIBUTOR CAP
3. REMOVE ROTOR
4. REMOVE IGNITION COIL DUST COVER



### Ignition Coil

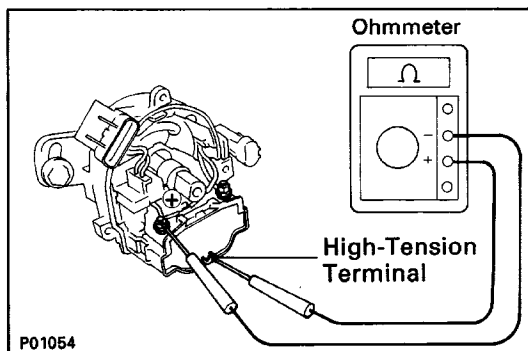
#### 5. INSPECT PRIMARY COIL RESISTANCE

Using an ohmmeter, measure the resistance between positive (+) and negative (-) terminals.

**Primary coil resistance (Cold):**

**0.3 ~ 0.6 at -10 ~ +40°C (14 ~ 104°F)**

If the resistance is not as specified, replace the ignition coil.



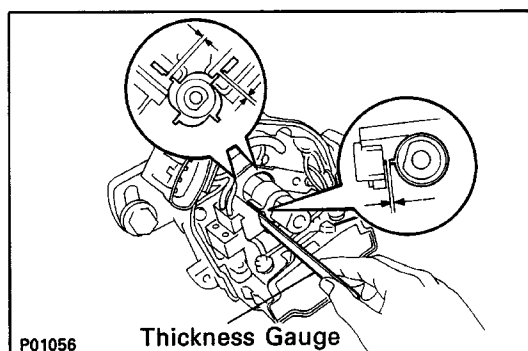
#### 6. INSPECT SECONDARY COIL RESISTANCE

Using an ohmmeter, measure the resistance between positive (+) and high-tension terminals.

**Secondary coil resistance (Cold):**

**9 ~ 15 k at -10 ~ +40°C (14 ~ 104°F)**

If the resistance is not as specified, replace the ignition coil.



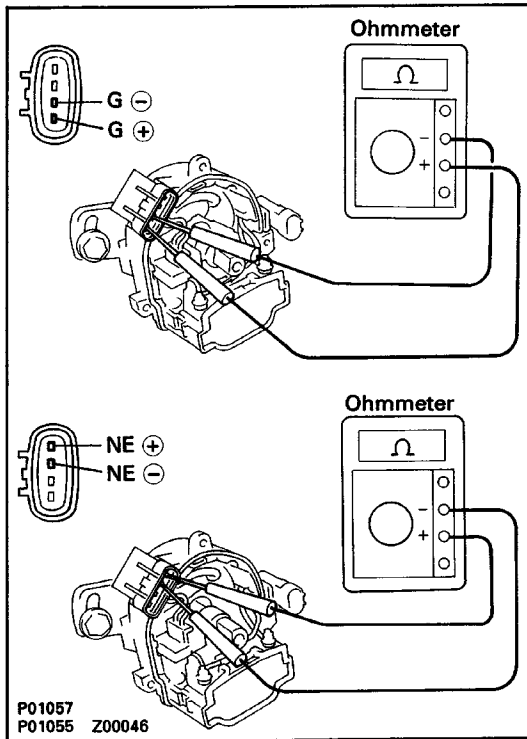
### Distributor

#### 7. INSPECT AIR GAP

Using a thickness gauge, measure the air gap between the signal rotor and pickup coil projection.

**Air gap: 0.2 ~ 0.4 mm (0.008 ~ 0.016 in.)**

If the air gap is not as specified, replace the distributor housing assembly.



## 8. INSPECT SIGNAL GENERATOR (PICKUP COIL) RESISTANCE

Using an ohmmeter, measure the resistance between the terminals (G (+) and G (-) , NE (+) and NE (-) ).

### Pickup coil resistance (Cold):

G (+) and G (-)

185 ~ 265 at -10 ~ +40°C (14 ~ 104°F)

NE (+) and NE (-)

370 ~ 530 at -10 ~ +40°C (14 ~ 104°F)

If the resistance is not as specified, replace the distributor housing assembly.

9. REINSTALL IGNITION COIL DUST COVER
10. REINSTALL ROTOR
11. REINSTALL DISTRIBUTOR CAP
12. RECONNECT DISTRIBUTOR CONNECTORS

## INSPECTION OF IGNITER

(See Spark Test procedure on page [IG-15](#))